

# Jubair Ahmed Nabin

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## Summary

Lecturer and researcher in Computer Science & Engineering at IUBAT, Dhaka. Research focuses on federated learning, intrusion detection systems, explainable AI (SHAP/LIME), adversarial machine learning, and video anomaly detection. Author of 7 peer-reviewed papers at IEEE and Springer venues.

## Education

**B.Sc. Chittagong University of Engineering & Technology (CUET)**, Computer Science and Engineering Sept 2018 – June 2023

- CGPA: 3.64 / 4.00
- Relevant Coursework: Machine Learning, Data Structures & Algorithms, Computer Networks, Database Management Systems, Artificial Intelligence, Software Engineering

## Experience

**International University of Business Agriculture and Technology (IUBAT)**, Lecturer, Department of CSE Dhaka, Bangladesh  
Jan 2023 – present  
3 years 9 months

- Courses taught: Artificial Intelligence, Data Science, Data Structures, Database Management Systems, System Analysis & Design
- Mentored 35+ undergraduate students on ML and cybersecurity research projects; two student-led projects advanced to peer-reviewed publication (IEEE PECCII 2026, IEEE ICMI 2026)

**CUET / IUBAT**, Researcher — Machine Learning & Cybersecurity Dhaka, Bangladesh  
Jan 2022 – present  
4 years 9 months

- Developed a privacy-preserving CNN-LSTM architecture within a Federated Learning framework for network intrusion detection, achieving over 99.36% accuracy across all seven attack classes on CIC-IDS2017 without centralizing raw data; raised F1-score on the hardest class (Web Attack) from 0.20 (CNN baseline) to 0.93, outperforming LSTM, CNN+BiLSTM, and CNN+GRU variants
- Built a novel file-entropy dataset and detection pipeline for crypto-ransomware, developing a custom FUSE-based filesystem to intercept and log file write/delete operations across 123 file types (30 ransomware strains, 93 benign); trained ML models achieving up to 96% accuracy (first-author, Springer LNNS 2025)
- Evaluated cross-domain adversarial robustness and XAI explanation stability in phishing detection, comparing PGD and Carlini-Wagner attack accuracy alongside SHAP explanation drift across feature-engineered vs. raw-URL input representations (IEEE PECCII 2026)
- Designed a two-stage CNN + Vision Transformer pipeline for video anomaly detection on the UCF-Crime dataset, reaching 98% accuracy on binary classification and 95% on multi-class categorization; introduced a keyframe extraction algorithm cutting per-video processing time to 20ms

## Publications

**Cross-Domain Adversarial Robustness and Explainability Stability in Phishing Detection Systems** June 2026

Md. Rifat Hassan, **J. A. Nabin**, Toyeer-E-Ferdoush  
[10.1109/PECCII70991.2026.11661863](https://doi.org/10.1109/PECCII70991.2026.11661863) (IEEE International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII))

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| <b>Unraveling the Factors of Procrastination: An Insightful Assessment of Procrastination Levels Through ML and XAI</b><br>Mariam Sarker, Anwar Hossain Efat, Maria Afrin Khan, Minhaz Zibran, <b>J. A. Nabin</b><br><a href="#">10.1109/ICMI68585.2026.11539948</a> (IEEE 5th International Conference on Computing and Machine Intelligence (ICMI))                                         | Apr 2026  |
| <b>A Novel File Entropy Dataset for Crypto-Ransomware Detection Using Machine Learning</b><br><b>J. A. Nabin</b> , Md Mokammel Haque<br><a href="#">10.1007/978-981-96-2721-9_20</a> (2nd International Conference on Machine Intelligence and Emerging Technologies (Springer LNNS))                                                                                                         | July 2025 |
| <b>Federated Deep Learning for Cybersecurity and Intrusion Detection in Decentralized Networks</b><br>Naeem Mia, <b>J. A. Nabin</b> , Suzad Mohammad, Mahedi Hasan, Fahim Shakil Tamim, Dipta Mohon Das<br><a href="#">10.1109/ICAECA63854.2025.11012607</a> (IEEE 3rd International Conference on Advancements in Electrical, Electronics, Communication, Computing and Automation (ICAECA)) | July 2025 |
| <b>A Novel Temporal-Aware Adaptive Feature Selection Strategy for Network Intrusion Detection Systems</b><br>Naeem Mia, Md Mahfuzul Haque Gazi, <b>J. A. Nabin</b> , Fahim Shakil Tamim, Suzad Mohammad, Md Rownak Ul Islam<br><a href="#">10.1109/GINOTECH63460.2025.11076711</a> (IEEE Global Conference in Emerging Technology (GINOTECH))                                                 | May 2025  |
| <b>Two-Stage Vision Transformer-Based Framework for Anomaly Detection and Classification in Surveillance Videos</b><br>Mahedi Hasan, <b>J. A. Nabin</b> , Naeem Mia, Fahim Shakil Tamim, Suzad Mohammad, Dipta Mohon Das<br><a href="#">10.1109/ECCE64574.2025.11013374</a> (IEEE International Conference on Electrical, Computer and Communication Engineering (ECCE))                      | Feb 2025  |

## Projects

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| <b>FL-IDS Framework</b> <ul style="list-style-type: none"> <li>Privacy-preserving CNN-LSTM federated learning architecture for network intrusion detection across distributed clients without raw data sharing</li> <li>Evaluated on CIC-IDS2017 across seven attack classes; 99.36% accuracy, Web Attack F1 raised from 0.20 to 0.93</li> </ul>             | 2022 – Present |
| <b>Crypto-Ransomware Detection Pipeline</b> <ul style="list-style-type: none"> <li>Built a FUSE-based filesystem to capture file entropy signals across 123 file types (30 ransomware families, 93 benign categories)</li> <li>Trained ML classifiers achieving up to 96% detection accuracy; published first-author at Springer LNNS 2025</li> </ul>        | 2023 – 2025    |
| <b>Adversarial Phishing Detection Study</b> <ul style="list-style-type: none"> <li>Measured SHAP explanation drift under PGD and Carlini-Wagner adversarial perturbations in phishing URL classifiers</li> <li>Quantified cross-domain robustness gaps across feature-engineered vs. raw-URL input representations; published at IEEE PECCII 2026</li> </ul> | 2024 – 2026    |

## Video Anomaly Detection System

2023 – 2025

- Two-stage CNN + Vision Transformer pipeline for spatial-temporal anomaly detection on the UCF-Crime dataset
- Introduced a keyframe extraction algorithm reducing per-video processing time to 20ms; 98% binary and 95% multi-class accuracy

## Skills

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**Programming Languages:** Python, C++, SQL, JavaScript

**ML / DL Frameworks:** PyTorch, TensorFlow, Keras, Scikit-learn

**Federated Learning:** FL aggregation pipelines, privacy-preserving distributed training

**Explainability & Security:** SHAP, LIME, adversarial robustness (PGD, Carlini-Wagner)

**Data & Visualization:** Pandas, NumPy, Matplotlib, OpenCV

**Tools & Environments:** Git, Linux, LaTeX, Jupyter Notebook, VS Code, Google Colab

## Awards

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**Research Publication Recognition:** Miyan Publications Reward Ceremony 2026

**Competitive Programming:** CUET Talent Hunt IUPC — 5th Position (2018–2022)

**Competitive Programming:** Meta Hacker Cup — Round 1 Qualifier (2020, 2022)

**Academic Merit:** General Board Scholarship — 2015, 2017

## Service

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**Conference Presenter:** IEEE PECCII 2026, IEEE ICMI 2026, IEEE ICAECA 2025, IEEE GINOTECH 2025, IEEE ECCE 2025, Springer MIET 2025